



Enhancing goal attainment in higher education with a scripted conversational agent: Effects of monitoring and reflection support in digital learning

Gabrielle Martins van Jaarsveld^{a,*}, Jacqueline Wong^b, Martine Baars^c,
Marcus Specht^{d,e}, Fred Paas^{a,f}

^a Department of Psychology, Education and Child Studies, Erasmus University Rotterdam, Rotterdam, Netherlands

^b Department of Education, Utrecht University, Utrecht, Netherlands

^c Behavioral Science Institute, Radboud University, Nijmegen, Netherlands

^d CATALPA, Fernuniversität in Hagen, Hagen, Germany

^e Faculty of Electrical Engineering, Mathematics and Computer Science, Delft University of Technology, Delft, Netherlands

^f School of Education, University of New South Wales, Sydney, Australia

ARTICLE INFO

Keywords:

Self-regulated learning
Goal setting
Goal attainment
Technology-enhanced learning
Conversational agents
Higher education

ABSTRACT

Self-regulated learning (SRL) is essential for academic success in higher education, yet many students struggle to effectively regulate their own learning behaviours. While goal-setting interventions can help students set high-quality goals as the foundation for their learning behaviours, additional supports are needed to help transition from goal setting into effective goal striving. This study examines the impact of monitoring and reflection supports, delivered through a scripted conversational agent, on students' goal attainment, SRL skills, and academic performance. In this study, 84 undergraduate students were randomly assigned to one of four experimental conditions: Control, Monitoring Only, Reflection Only, or Monitoring & Reflection. Over a five-week intervention, participants engaged in weekly goal-setting activities, with additional monitoring and/or reflection prompts depending on their assigned condition. Results showed that participants in the Monitoring Only and Monitoring & Reflection conditions reported significantly higher goal attainment than those in the Reflection Only and Control groups, suggesting that monitoring plays a critical role in reinforcing goal-directed behaviour. While SRL skills improved across all conditions, no significant differences were found between groups, indicating that consistent goal setting alone may support SRL development. There were no significant effects of the intervention on academic performance. These findings highlight the immediate effectiveness of progress-monitoring activities and suggest that reflection may require longer intervention periods to have significant effects. This study supports the use of conversational agents for delivering scalable SRL support and provides insights into the design of multi-phase SRL supports in digital education.

Self-regulated learning (SRL) skills are crucial for students to succeed in higher education (Richardson et al., 2012). Students in higher education are expected to drive their own learning processes, often with limited oversight from instructors. Digital learning

* Corresponding author.

E-mail address: martinsvanjaarsveld@essb.eur.nl (G. Martins van Jaarsveld).

environments that are increasingly common in higher education amplify this challenge, as students must self-regulate their learning with minimal real-time instructor feedback, necessitating scalable tools like conversational agents to support SRL (Winne and Hadwin, 2008). In such contexts, structured support for goal monitoring and reflection becomes critical in sustaining motivation and ensuring progress toward learning objectives. As Wong (2013) notes, higher education expects students to become 'self-managing and self-directed learners' (p. 130), placing significant demands on their SRL skills. Students who have not developed robust SRL skills often face challenges in setting goals, selecting and implementing appropriate learning strategies, and reflecting on their progress as they work towards those goals (Panadero, 2017; Zimmerman, 1990).

SRL is a cyclical process in which students are behaviourally and metacognitively active in their own learning, setting academic goals and employing planning, monitoring, learning, and reflection strategies to achieve them (e.g., Zimmerman, 1990). Several models have been created to explain SRL (Panadero, 2017), but Zimmerman's model (Zimmerman, 2000) has been particularly influential and focuses on the three core phases that are also present in other SRL models (Panadero, 2017). These phases include 1) forethought, where students set goals and plan, 2) performance, where they implement strategies and monitor progress, 3) self-reflection, where they evaluate efforts and outcomes (Zimmerman, 2000). These phases form a cyclical process in which each new cycle is shaped by the insights gained from the previous phase.

Prior studies have shown that students often run into problems during all phases of the SRL process, including struggling to set effective goals (Kochekseraii, 2019; Nurjannah et al., 2020), failing to implement effective learning strategies (Broadbent & Poon, 2015; Kizilcec et al., 2017), inaccurate progress monitoring (Guo, 2022), and difficulty with self-evaluation and reflection (Winkel et al., 2017). As a result, supporting SRL has become an important area of research, with many studies seeking to understand how to best support SRL (Wong et al., 2019). However, this has proven to be a complex topic with prior reviews finding mixed results as to how best to approach it (Heikkinen et al., 2023; Wong et al., 2019). Thus, this study explores how a scripted conversational agent, as a digital intervention, supports the monitoring and reflection phases of SRL to enhance goal attainment, SRL skills, and academic performance in higher education. By testing the individual and combined effects of monitoring and reflection within a structured goal-setting intervention, this research provides insights into which strategies are most effective in translating goal intentions into sustained self-regulatory behaviours and academic success.

1. Goal setting and SRL

In digital learning environments, effective SRL relies on many sub-processes, driving transitions between the forethought, performance, and reflection phases. As a goal-driven process, SRL hinges on effective goal setting, which underpins strategy selection, monitoring, and reflection. As a result, ineffective goal setting can lead to a misattribution of effort throughout the following stages and result in poor learning outcomes (Liem et al., 2008; Świątkowski & Dompnier, 2021). Therefore, goal-setting interventions are frequently examined as a form of SRL support within higher education settings (Martins van Jaarsveld et al., 2025b). These interventions aim to intercede early in students' SRL process, providing support to improve the goals which are set, and thus provide a strong base for the SRL phases which follow.

Studies have found that goal-setting interventions can be a highly effective means of improving learning and academic performance (Martins van Jaarsveld et al., 2025b). Nevertheless, prior research indicates that high-quality goals do not always translate into improved goal attainment (Martins van Jaarsveld et al., 2025a). While interventions targeting goal setting can help students generate clearer or more specific goals, they may not be sufficient to ensure the successful execution of those goals. Goal attainment can be hindered by failures in self-monitoring and lack of reflection on progress (Locke & Latham, 2019). Without active engagement in these SRL phases, students may abandon goals prematurely or fail to adapt their strategies to overcome challenges. Locke and Latham (2019) claimed that without ongoing feedback, monitoring, and adaptive strategy use, the initial benefits of setting high-quality goals can wear off over time. In support of this, Martins van Jaarsveld et al. (2025a) studied the effect of a goal-setting intervention carried out in a higher education population. In this study, the effect of prompting and feedback during a goal-setting intervention were examined. The results found a significant effect of both types of goal-setting supports on goal quality, but no effect on goal attainment; thus, a goal-setting intervention which improved the quality of the goals students set did not improve their chances of successfully attaining those goals. This was partly due to the intervention's failure to enhance goal-striving behaviours, leaving students with high-quality goals no more likely to pursue them between sessions. These findings suggest that the disconnect between goal setting and goal attainment may stem from challenges in transitioning from the metacognitive process of goal setting to the behavioural goal striving. Thus, goal setting alone can fail to induce sufficient goal commitment to drive any further action (Oettingen et al., 2001). A possible solution is to explore how the subsequent phases of the SRL cycle can be supported to close the gap between goal setting and goal attainment.

2. Monitoring and reflection as crucial components of SRL

Goal monitoring and reflection play a critical role in ensuring effective SRL, helping inform and drive goal striving behaviour (Panadero, 2017; Zimmerman, 2000). Goal monitoring involves the tracking of progress towards a specific goal or outcome and serves as an important mechanism for sustaining motivation and engagement during the learning process (Schunk & DiBenedetto, 2020). Reflection is the act of reviewing and evaluating one's goal striving behaviours after working towards the goal, noting effective and ineffective learning strategies and using this knowledge as the basis for adaptations for future behaviour (Zimmerman, 1998). Prior studies have shown that both goal monitoring and reflection can have a significant positive effect on goal attainment and support the translation of intentions into concrete goal-directed behaviours (Guo, 2022; Winkel et al., 2017). Without support for monitoring and

reflection, students may struggle to maintain focus on their academic objectives, effectively manage their learning strategies, or adapt their approaches based on goal progress.

2.1. The role of goal monitoring in SRL

Monitoring supports the generation of internal feedback, increasing awareness about the discrepancies between the effort and behaviour currently being carried out, and that needed to attain the goal (Griffin et al., 2013). By consistently assessing progress and identifying discrepancies between actual performance and goal expectations, students can make real-time adjustments to their strategies (Schunk & DiBenedetto, 2020). This adaptive self-regulation ensures sustained effort and enhances goal attainment over time. Monitoring therefore increases goal directed behaviours and awareness about the learning activities contributing towards goal attainment and forms a crucial element in supporting the transition from goal setting into goal striving. Interventions targeting monitoring generally take the form of check-in prompts, or learning diaries, which scaffold progress monitoring, often in a structured and quantitative manner (Guo, 2022). Meta-analyses exploring the effect of goal monitoring on goal attainment both inside the educational field (Guo, 2022) and more broadly (Harkin et al., 2016) report that goal monitoring is generally found to have small to medium positive effects on goal attainment, and that monitoring interventions are an effective means of improving performance. For example, Scheithauer and Kelley (2017), found that self-monitoring instruction provided to higher education students with ADHD significantly improved academic performance and goal attainment, while Dignath-van Ewijk et al. (2015) found that a self-monitoring intervention delivered using a learning diary significantly improved metacognitive attitude and estimated learning efficacy.

2.2. The role of reflection in SRL

Reflection fosters improvements in SRL behaviours across successive SRL cycles, leveraging insights from the current goal-setting experience to enhance future goals, whereas monitoring primarily influences the achievement of the immediate goal by influencing goal-striving behaviours in the current cycle. (Griffin et al., 2013; Zimmerman & Moylan, 2009). Reflection supports student awareness about their learning activities and supports the improvement in goal and strategy selection, especially with repeated reflective prompting (Zarei Hajiabadi et al., 2023; Zhai et al., 2023). Interventions targeting reflection generally focus on prompting self-reflection in learning contexts using writing exercises or dialogues to provide scaffolds that guide students through the process (Winkel et al., 2017). A meta-analysis by Winkel et al. (2017) exploring the use of reflection activities in medical education found that they significantly increased learning, especially of complex topics, and influenced a range of motivational, cognitive, and meta-cognitive processes during the learning process. For example, Choi et al. (2023) implemented reflective prompts combined with hints in an online programming course, boosting task performance significantly, while Bannert et al. (2015) used digital metacognitive scaffolds to enhance long-term SRL gains.

2.3. Supporting the three cyclical phases of SRL

Prior research suggests that supports targeting multiple phases of the SRL cycle are more effective than those offering support for only a single phase (Guo, 2022; Wong et al., 2019). Goal setting, monitoring, and reflection are complementary processes, where goal setting informs and directs the application of attention, monitoring serves to improve awareness of progress and trigger the generation of internal feedback, and reflection prompts metacognitive processes encouraging learners to critically evaluate their strategies and make adaptive improvements (Hattie & Timperley, 2007; Panadero, 2017). When combined, they create a reinforcing cycle of self-regulation, allowing students to engage in a more effective approach to learning (Zimmerman & Moylan, 2009). Therefore, incorporating goal setting, monitoring, and reflection support within SRL interventions is likely to maximize SRL skills, goal attainment, and academic performance improvements, exceeding the benefits of single phase supports.

3. Scripted conversational agents as SRL support tools

While research has shown the importance of supporting SRL, it also highlights the complexity of doing so effectively, and studies suggest that SRL development requires a multi-faceted and holistic approach (Panadero, 2017). Conversational agents offer a scalable, structured, and adaptive way of delivering support in digital learning environments. Conversational agents are programs designed to simulate natural language conversation, and thus allow for the creation of interactive, dialogue-based interventions that can guide students through SRL processes in a personalized yet automated manner. Conversational agents have been increasingly applied in higher education contexts and show promise in supporting SRL by delivering timely prompts, feedback, and structured reflection opportunities (Guan et al., 2025). One of their key advantages lies in their ability to initiate and sustain engagement without relying on participants to drive the process. This is particularly valuable given that learners most in need of SRL support may also be the least likely to seek out and make use of support tools. A meta-analysis by Wang et al. (2021) found that externally guided SRL interventions led by researchers were significantly more effective than those led by participants. Conversational agents therefore serve as a delivery mechanism as well as an active facilitator of SRL.

Much of the existing literature conflates scripted, rule-based agents with generative, AI-driven chatbots, which makes it impossible to judge the incremental learning value of artificial intelligence. Establishing the effectiveness of a minimal, non-generative baseline is therefore a necessary first step. The present study deliberately employs such a baseline so that future research can quantify what, if anything, adaptive or generative intelligence adds.

4. The current study

Since improving goal quality in goal-setting interventions does not always translate to goal attainment (Martins van Jaarsveld et al., 2025a), we aimed to close this gap by supporting the monitoring and reflection phases of the goal setting cycle in the current study. We tested the effect of goal monitoring and reflection activities, aiming to support the transformation of high-quality goals into goal-directed behaviours, increasing the likelihood of goal attainment and subsequently improving performance. This approach is supported by earlier reviews which suggest that SRL interventions focusing on all stages of the SRL cycle are more effective than those targeting only a single phase (e.g., Guo, 2022; Wong et al., 2019).

This study deployed a text-based conversational agent, built on a scalable platform, to deliver a goal-setting intervention to higher education students, adapting the setup from Martins van Jaarsveld et al. (2025a). Conversational agents provide a scalable and structured method for guiding students through SRL processes while minimizing instructor workload. Prior research suggests that AI-driven tutoring systems can enhance SRL behaviours by providing adaptive feedback and promoting metacognitive engagement (Azevedo et al., 2010). To examine the impact of goal monitoring and reflection in a goal-setting intervention on goal attainment, SRL skills, and academic performance, the study employed four experimental conditions, including Control (CT), Monitoring Only (MO), Reflection Only (RF), Monitoring & Reflection (MO + RF). With this study, we aim to answer the following research question:

What is the effect of goal monitoring and reflection, individually and combined, on the attainment of self-set academic goals, SRL skills, and academic performance?

To answer this research question, we tested three sets of hypotheses, with each set relating to one of the main outcome measures. The first set of hypotheses relate to the effect of goal monitoring and reflection on goal attainment. Based on the findings of prior research (e.g., Guo, 2022), we expected monitoring to have an immediate positive effect on goal attainment, reflection to have a delayed but cumulative positive effect, and for support of both phases to be complementary and have the largest impact when combined. Therefore, we hypothesized a significant main effect of condition, with MO + RF resulting in the highest goal attainment, followed by MO and RF which would not be significantly different from each other, followed by CT with the lowest goal attainment (H1a). Secondly, we hypothesized that there would be a significant main effect of time (H1b), and a significant interaction effect between condition and time, with MO + RF, MO, and RF groups all showing significant increases over time, but CT remaining stable across the 5-weeks (H1c).

The second set of hypotheses relate to the effect of goal monitoring and reflection on SRL skills. Based on the findings of prior studies (e.g., Guo, 2022; Wong et al., 2019), we expected that repeated prompting of both monitoring and reflection phases would improve the quality of that SRL process, thus resulting in an overall increase in SRL skills, with support for multiple phases resulting in a larger improvement to overall SRL skills. In the case of this study, all conditions were supported during the goal setting process, with monitoring groups targeting monitoring processes, and reflection groups targeting self-reflection processes. This means that the CT condition received support for a single phase of the SRL cycle, with the MO and RF receiving support for two phases of the SRL cycle, and the MO + RF condition receiving support across the whole SRL cycle. Therefore, we hypothesized a significant main effect of condition (H2a), as well as a significant main effect of time with a general improvement in SRL skills over time for all conditions (H2b). We also hypothesized a significant interaction effect, with the MO + RF condition increasing the most over time, followed by the MO and RF conditions with the CT condition increasing the least (H2c).

The final set of hypotheses concern the effects of goal monitoring and reflection on academic performance. We expected that the intervention would improve academic performance by targeting goal attainment and SRL skills. Prior studies have found that both monitoring and reflection interventions can have a positive effect on academic performance, both when they are addressed separately (Guo, 2022; Winkel et al., 2017) or combined (Heikkinen et al., 2023; Wong et al., 2019). Therefore, we hypothesized a significant difference in post-intervention academic performance between conditions, with all three active conditions having significantly better academic performance than the CT condition (H3a).

5. Method

5.1. Study design

This study employed a randomized controlled trial design, in which participants were randomly allocated to one of four study conditions. An a-priori power analysis using G*Power v3.1 (Erdfelder et al., 2009; Faul et al., 2007) was carried out for the repeated measures ANOVA to test within-between interactions, with four groups, and four measures. Based on prior studies exploring the effect of goal-setting interventions (Martins van Jaarsveld et al., 2025a; Wang et al., 2021) and monitoring interventions (Guo, 2022) on goal attainment, SRL, and performance, we expected medium effect sizes. Therefore, the effect size was set at $F = 0.25$ (Cohen, 1988), $\alpha = .05$. A previous study using a similar setup and conversational agent found correlation among repeated measures to be between 0.250 and 0.350 and thus the mid-point of 0.300 was used for the power analysis. A final recommended minimum sample size of 72 was calculated for an actual power of 0.96. Given the expected 20–30 % attrition rate and based on similar studies previously carried out (e.g., Martins van Jaarsveld et al., 2025a), Martins van Jaarsveld et al., 2025b final sample of 100 participants was selected to ensure a sufficient sample size for the final analysis. Ethical approval was obtained from the ethics committee prior to the start of the study at the institution where data collection was carried out (ETH2223-0441).

5.2. Participants

Participants were 102 bachelor students from the social sciences faculty at a large Dutch university. Participants voluntarily signed up to participate in the study, which was completed during April–May 2024. Participants enrolled via the university's online research portal and received three research credit hours in exchange for their participation. These research credits counted towards the total of 20 h that all students in the department are required to complete.

Of the 102 participants, 5 participants signed up but did not respond to any further communication regarding the study. An additional 13 students had incomplete datasets, having started the study but missing more than one activity during the 5-weeks. Of the remaining participants, four indicated in the closing questionnaire of the study that they had used a large language model (LLM) to generate some of their responses. Given that the use of LLM could have introduced uncertainty regarding the extent of students' independent participation and engagement with the activities of the study, we conducted all main analyses both with and without these participants to check. As their inclusion did not lead to any significant changes in the results, we retained them in all final reported analyses. This resulted in a final sample of 84 students. This sample consisted of primarily first year bachelor students ($n = 57$, 67.9 %), with the remainder of participants coming from the second ($n = 13$, 15.5 %) and third ($n = 14$, 16.7 %) year. Participants' ages ranged from 18 to 26 with an average age of 20.38 ($SD = 1.78$), and the majority female sample ($n = 67$, 79.8 %) reflected the uneven gender distribution of the program.

5.3. Materials and measurements

5.3.1. Guided goal setting, goal monitoring, and reflection with a scripted conversational agent

The intervention was carried out by a text-based conversational agent using a pre-designed script. This agent was developed using Typebot, an open source chatbot builder (Arnaud, 2024). An overview of the conversational agent script for the intervention activities, as well as the scoring and feedback workflow can be found in the supplementary materials.

Conversational Agent Design. The conversational agent was designed as a fully scripted, rule-based interaction, with the core of the interactions delivered being identical within conditions. The prompts, agent responses, and branching logic used in the intervention were pre-programmed by researchers prior to the start of the study. This conversational agent did not contain any generative AI components, with all conversational elements being scripted by researchers beforehand.

Conversational Agent Architecture. The scripted agent is implemented as a finite-state network with 73 dialogue nodes that cover all weekly activities (see S1a and S1b in Supplementary Materials). Every system prompt, branch condition, and feedback message was authored before the study. At run time the agent simply traverses the predefined graph according to if-then rules and never generates new text. No machine-learning, predictive modelling, or generative components are involved. This architecture guarantees identical wording across participants while still allowing the agent to select feedback that aligns with the goal-quality scores entered by the researchers.

Goal-Setting Activity. The goal-setting activity took place weekly from the first week of the study on Monday. The setup was similar to Martins van Jaarsveld et al. (2025a). During the goal-setting activity, participants in all four conditions were asked to set a goal using the prompt "Set a goal related to your studies which you would like to achieve in the coming week." to set goals. Participants were then guided through the process of adding additional details to their goal including 1) more specific details about what the goal entails, 2) how they plan to measure the goal, 3) why the goal is important to them, and 4) a multistep plan to achieve the goal in the upcoming week. These prompts were context independent, and did not mention any specific course or study activity in which to ground the goals, thus allowing participants the autonomy to set goals related to any relevant study activities.

Each week the quality of all goals was manually scored by one of the researchers using a rubric adapted from Lockspeiser et al. (2013), with a score of 0–2 being assigned for four characteristics: specificity, measurability, personal importance, and multisource planning. Scores were input into the backend of the conversational agent setup, and the agent then selected appropriate feedback from a selection of pre-determined feedback scripts. Feedback was provided on the characteristic which the participant scored lowest on the previous week in all subsequent weeks. This process allowed for a rule-based form of adaptation, with the agent dynamically adjusting feedback on the basis of goal quality scores provided by researchers, but not generating any new responses, allowing for a standardized and consistent interaction across all participants. For each week, 10 goals were scored independently by a second researcher. The week-to-week Cohen's kappa measure of interrater reliability ranged from 0.781 to 0.921, for an average Cohen's kappa of $\kappa = 0.842$.

Goal Monitoring Activity. The monitoring activity took place weekly from the first week of the study on Thursday/Friday during the check-in session. During this activity, participants in the MO or MO + RF conditions were shown the goal they set earlier in the week and given a series of prompts asking them to discuss their progress on the goal, the work they had already done, and what they still needed to do to achieve the goal. A series of three prompts were designed based on items pertaining to progress monitoring from the metacognition subscale of the Self-Regulated Online Learning Questionnaire (SOL-Q; Jansen et al., 2017).

Reflection Activity. The reflection activity was implemented weekly from week 2 of the study for the RO and MO + RF conditions, during the Monday/Tuesday session, before the next round of goal setting. During this activity participants were given a series of three prompts asking them to discuss learning strategies which had gone well the previous week, strategies which had gone poorly the previous week, and what they had learned which could be used to improve future iterations of goal setting and striving. Prompts were designed based on items pertaining to self-reflection and adaptation from the metacognition subscale of the SOL-Q (Jansen et al., 2017).

5.3.2. Goal attainment

Goal attainment was measured using a weekly self-report question each Monday before participants set a new goal. Participants were first reminded of the goal they set the previous week and asked to score the extent to which they felt they achieved the goal in the previous week on a 5-point Likert scale ranging from “Not achieved at all” to “Achieved it fully”. The goal attainment measure for the final goal of the experiment fell outside of the 5-week experimental period, as such, goal attainment was measured for the first 4 goals, although participants set five goals in total.

5.3.3. SRL (pre & post)

SRL was measured using the Online Self-regulated Learning Questionnaire (OSLQ; Barnard et al., 2009). Specifically, we adopted four of the six subscales: goal setting (5-items), task strategies (4-items), time management (3-items), and self-evaluation (4-items). The environment structuring and help seeking subscales were not measured as these processes were not addressed in the intervention. We adjusted the phrasing of a few items to fit the primarily blended nature of students’ education. Participants were asked to rate the extent to which they agree with each item on a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The average of all 16 items was taken, and thus a final SRL score between 1 and 5 was calculated for each participant at both pre-and post-intervention moments. The Cronbach’s alpha for this SRL measure was $\alpha = .785$ for the pre-intervention measure, and $\alpha = .784$ for the post-intervention measure, indicating good levels of internal consistency.

5.3.4. Academic performance

Academic performance was measured using official exam grades provided by the university. The grading system follows a 10-point scale, with students receiving a grade for each completed course. Each course is assigned a specific number of credits based on its workload, allowing for a weighted calculation of academic performance. The academic year at the university is divided into semesters, referred to as blocks, which vary in length depending on the student’s year of study. Blocks are either 5 or 8 weeks long, with students completing multiple courses per block. There were differences in the number of courses and credits being taken by participants at the time of the study, with differences existing based on study year, and on the course load of individual students. To calculate academic performance, a weighted semester average was calculated by averaging course grades while adjusting for the corresponding course credits. The study started in week 1 of a block, ensuring that it covered the entire duration of a 5-week block and the majority of an 8-week block. Final exams for each block took place at the end of the final week of that block. As a result, pre-intervention academic performance was measured using the semester immediately preceding the study and was used to check for baseline differences between conditions. While post-intervention academic performance was assessed using grades from the semester during which participants participated in the study and was used to test the effect of the intervention on performance.

5.4. Procedure

At the start of the study, all students who had registered to participate received an email containing practical information about the study and a link to an online questionnaire administered via Qualtrics. This questionnaire included the informed consent form and the pre-intervention SRL measure. Upon providing consent and completing the initial survey, participants were randomly assigned to one of four experimental conditions: CT, RF, MO, or MO + RF. They were then redirected to the conversational agent to complete the initial goal-setting activity. The intervention spanned five weeks, during which participants engaged in a series of prompted SRL activities based on their assigned condition and delivered via the conversational agent. All participants received an email every Monday

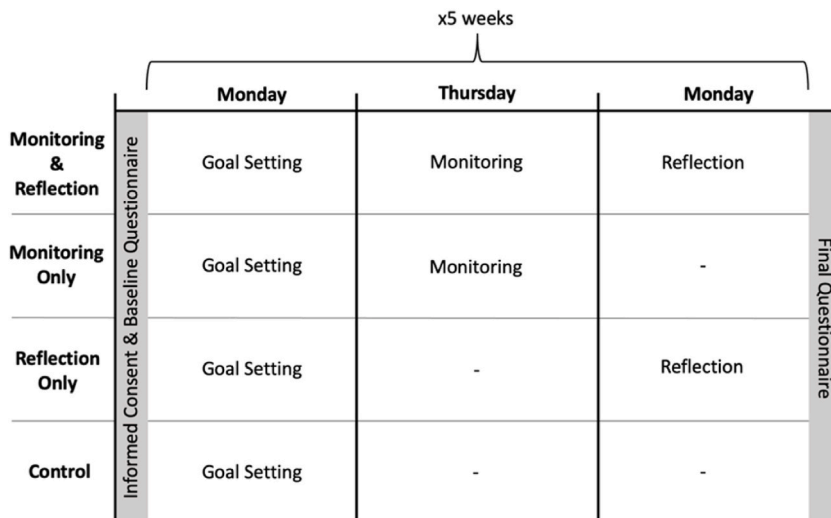


Fig. 1. Overview of study procedure with conversational agent interactions completed during the 5-weeks across the four conditions.

prompting them to access the conversational agent and complete the goal-setting activity. Participants assigned to the MO and MO + RF conditions received a separate email every Thursday, prompting them to complete a monitoring activity via the conversational agent. From week 2 of the study, all participants started the weekly goal-setting activity by first reporting their attainment of the goal that they had set in the previous week. Additionally, those in the RF and MO + RF conditions carried out a reflection activity before proceeding to setting a new goal. This cycle of goal setting, goal monitoring and reflection was repeated weekly for five weeks. After the final goal-setting session, all participants completed a concluding questionnaire via Qualtrics, which included the post-intervention SRL measure. Fig. 1 shows an overview of the weekly SRL activities completed by each of the study conditions.

5.5. Analysis

To test hypotheses H1a, H1b, and H1c, a mixed-design Analysis of Variance (ANOVA) was run. For this analysis, there was one continuous outcome variable, which was goal attainment, as well as two independent variables, a between-subjects factor consisting of the four treatment conditions, and a within-subjects factor which is time (4 measurement occasions).

To test hypotheses H2a, H2b, and H2c, a second mixed-design ANOVA was carried out. For this analysis, there was one continuous dependent variable which was SRL skills. There were also two independent variables, a between-subjects factor consisting of the four treatment conditions, and a within-subjects factor which was time (2 measurement occasions).

To test hypothesis H3a a one-way ANOVA was carried out. For this analysis, there was one continuous dependent variable which was post-intervention academic performance, and experimental condition as the independent variable. Prior to this analysis, an additional one-way ANOVA was performed to assess pre-intervention academic performance across groups, ensuring that no significant differences existed at baseline.

For all research questions, post-hoc tests were conducted to further explore significant findings, using Bonferroni corrections to compensate for multiple comparisons.

6. Results

6.1. Overview of outcome measures

The means and standard deviations for each outcome measure per condition can be found in Table 1.

6.2. Effect on goal attainment

All assumptions were met for conducting mixed-design ANOVA. The results showed a significant main effect of condition, $F(3, 80) = 10.728, p < .001, \eta_p^2 = .287$, indicating a large effect. Results of pairwise comparisons with Bonferroni corrections showed significant differences in goal attainment across conditions. Participants in the MO + RF condition achieved significantly higher scores compared to those in the RF condition ($p = .002$) and the CT condition ($p < .001$). Similarly, participants in the MO condition showed significantly higher goal attainment than those in the RF ($p = .031$) and CT conditions ($p < .001$). No significant difference was found between the MO + RF and MO conditions ($p > .05$), nor between the RF and CT conditions ($p > .05$). These results partially support hypothesis H1a, indicating that monitoring conditions (MO, MO + RF) outperformed non-monitoring conditions (RF, CT), while reflection alone yielded no advantage over the control. This suggests that real-time performance feedback and self-monitoring cues serve as critical reinforcers for goal striving. By contrast, reflection alone may require longer engagement periods before observable behavioural changes can be seen. Fig. 2 shows the difference in goal attainment over time, between conditions.

Additionally, there was no main effect of time on goal attainment, $F(3, 240) = 1.967, p = .120, \eta_p^2 = .024$, and the interaction between time and condition was not significant, $F(9, 240) = 0.835, p = .584, \eta_p^2 = .030$. As a result, no further follow analyses were carried out, and hypotheses H1b and H1c were not supported.

Table 1

Descriptive statistics for goal attainment, SRL skills, and academic performance per condition.

Outcome Measure	Control (CT) (<i>n</i> = 26)		Reflection Only (RF) (<i>n</i> = 24)		Monitoring Only (MO) (<i>n</i> = 16)		Monitoring & Reflection (MO + RF) (<i>n</i> = 18)	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Goal Attainment								
Goal 1	2.88	1.03	2.63	1.25	3.56	1.26	3.83	0.79
Goal 2	2.77	1.03	3.12	1.39	3.44	0.96	3.56	0.98
Goal 3	2.85	1.08	3.08	1.35	4.87	0.89	3.89	0.96
Goal 4	2.77	0.95	3.29	1.37	3.88	1.09	4.22	1.06
SRL Skills								
Pre	3.37	0.41	3.20	0.63	3.30	0.52	3.35	0.37
Post	3.55	0.41	3.40	0.58	3.34	0.52	3.49	0.38
Academic Performance								
Pre	6.84	1.23	6.98	0.64	6.83	1.78	6.65	1.51
Post	6.97	0.82	7.17	0.80	6.73	1.55	7.02	0.74

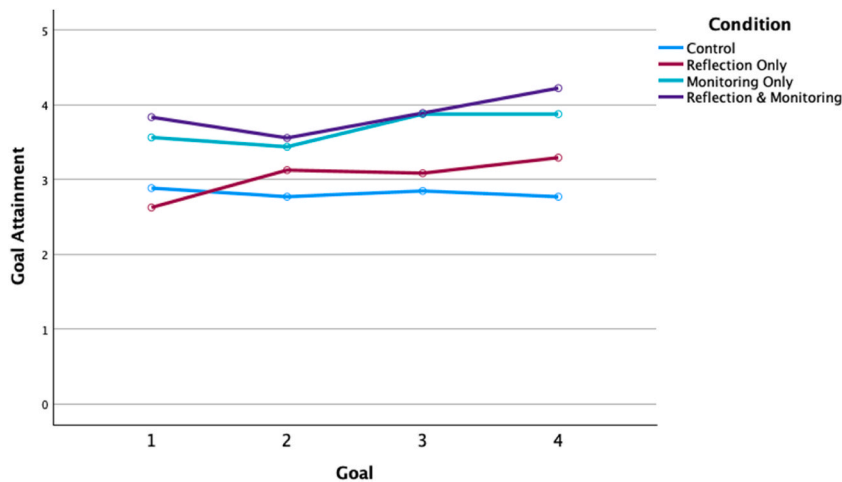


Fig. 2. Differences between conditions in changes in goal attainment over time.

6.3. Effect on SRL skills

All assumptions were met for mixed-design ANOVA. Results revealed a significant main effect of time, $F(1, 80) = 10.453, p = .002, \eta_p^2 = .116$, a moderate effect, indicating that SRL skills increased during the intervention. This supports hypothesis H2b. However, there was no significant main effect of condition, $F(3, 80) = 0.639, p = .592, \eta_p^2 = .023$, and the interaction between time and condition was not significant, $F(3, 80) = 0.607, p = .612, \eta_p^2 = .022$, meaning hypotheses H2a and H2c were not supported. Fig. 3 shows the changes in SRL from T1 to T2 between conditions. This figure supports the findings indicating that SRL skills did slightly increase across the intervention, however, this trend was the same for all conditions.

6.4. Effect on academic performance

To test hypotheses H3a, a one-way ANOVA was carried out, comparing the differences in academic performance between conditions. An initial test showed no significant differences in pre-intervention academic performance between groups, $F(3, 77) = 0.226, p = .878, \eta^2 = 0.009$, and thus the analysis was carried out.

The results showed no significant effect of condition on post-intervention academic performance, $F(3, 77) = 0.608, p = .612, \eta^2 = 0.024$, not supporting hypothesis H3a.

7. Discussion

SRL is a vital skill for students to succeed in higher education (Richardson et al., 2012), with the process being driven by goal setting (Panadero, 2017). However, both monitoring and reflection play an important role in the SRL cycle, supporting the transition from

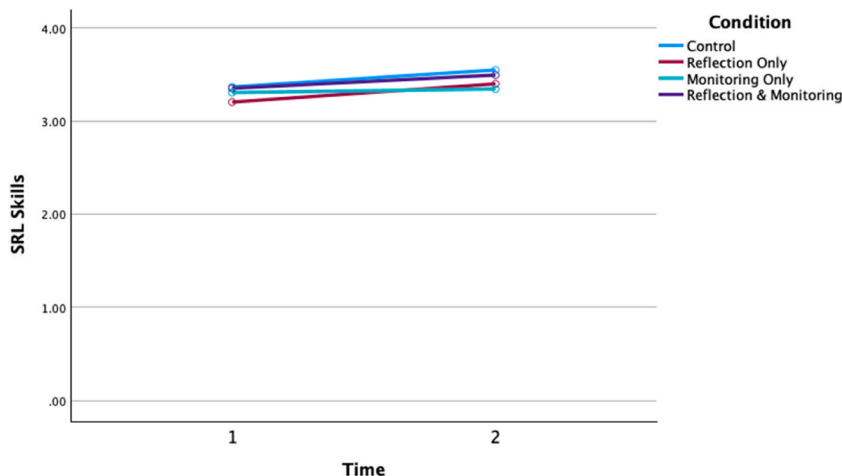


Fig. 3. Changes in SRL skills over time between conditions.

goal setting to goal striving (Panadero, 2017). In this study, we carried out a goal-setting intervention delivered via a conversational agent, and tested the effect of monitoring and reflection supports on SRL skills, goal attainment and performance. We aimed to answer the research question: *What is the effect of goal monitoring and reflection, individually and combined, on the attainment of self-set academic goals, SRL skills, and academic performance?* Our findings highlight monitoring's pivotal role in boosting goal attainment, with MO + RF and MO conditions outperforming RF and CT, suggesting that progress tracking through goal monitoring drives goal-directed behaviour. There was also a significant effect of time on SRL, with students in all conditions reporting improvement in SRL skills from pre-to post-intervention, but with no significant differences between conditions. There was no significant effect of the intervention on academic performance.

The first analysis tested the effect of monitoring and reflection on goal attainment. Hypothesis H1a was partially supported, as results showed that participants in the MO and MO + RF conditions reported significantly higher goal attainment compared to the RF and CT conditions. However, neither H1b or H1c were supported, as there was no significant effect of time on goal attainment, and no interaction effect between time and condition. The positive effect of monitoring was consistent with previous research (Guo, 2022; Harkin et al., 2016), where goal monitoring was found to significantly improve goal attainment, supporting the idea that progress tracking encourages goal-striving behaviours. This suggests that prompting monitoring was effective in supporting students in maintaining engagement with their academic goals and ensuring follow-through. The prompts seem to be successful in supporting students in identifying their current progress and adapting effort and focus for the remaining learning behaviours to ensure they could meet their goal. However, in contrast, the results showed that reflection alone did not result in significantly better goal attainment than the control condition, indicating that without additional support for progress tracking, reflection may not be sufficient to drive immediate behavioural modifications during goal striving. This finding aligns with prior studies suggesting that while reflection fosters long-term SRL development, its effects on immediate goal attainment may be more gradual and indirect (Bannert et al., 2015; Zhai et al., 2023), suggesting that reflection alone may require extended engagement before resulting in measurable behavioural changes. Alternatively, the weekly goal setup may have contributed to the lack of effect of reflection, as the goals changed weekly and what worked one week may no longer have been applicable the following week, and thus reflection could have a limited effect week-to-week. Overall, these findings challenge the assumption that reflection and monitoring function synergistically when combined in the short term (Zimmerman & Moylan, 2009), and suggests that monitoring may be the more immediate driver of goal attainment, whereas reflection's effect may depend on other factors such as duration or depth. Additionally, it is not clear the extent to which students are implementing the SRL behaviours prompted in the intervention, and this may also play a role in the findings. Future research could potentially further explore this, by adopting a learning analytics approach and embedding the conversational agent in a learning environment to collect behavioural data and thus be able to study SRL behaviours at a task level as well as offer support in real-time.

The second analysis tested the effect of monitoring and reflection on SRL skills. The results showed there was no significant effect of condition on SRL skills, and therefore hypothesis H2a was not supported. However, there was a significant effect of time on SRL skills, supporting hypothesis H2b. Additionally, the interaction effect between condition and time was not significant, and thus hypothesis H2c was not supported. These results indicate a general rise in SRL skills across all conditions, likely due to the consistent goal-setting practice inherent in the intervention, irrespective of additional monitoring or reflection support. This finding aligns with prior research suggesting that structured goal-setting activities alone promote SRL skill development (Zimmerman & Schunk, 2011). However, the lack of a condition-specific effect raises questions about whether monitoring and reflection require longer-term engagement to yield measurable differences in self-regulatory capacity. These findings would be in line with prior research showing that goal-setting activities can have a positive effect on SRL skills (Martins van Jaarsveld et al., 2025b), (Martins van Jaarsveld et al., 2025) but contrasting the expectation that monitoring and reflection would have an additional cumulative positive effect. Overall, this supports the use of goal-setting interventions to improve SRL skills, however, more research is needed to better understand how monitoring and reflection activities play a role in this process. Future research should focus on better understanding monitoring and reflection and could adopt a multi-method approach leveraging learning analytics and combining log data with self-report measures of SRL to get a more comprehensive understanding.

The third and final analysis tested the effect of monitoring and reflection on academic performance. The results showed no significant differences between groups in academic performance after the intervention, and thus hypothesis H3a was not supported. Some prior studies have found that SRL interventions can have a significant positive effect on academic performance (Heikkinen et al., 2023; Wong et al., 2019), however this is not consistently the case across all goal setting and SRL research, and findings, especially for shorter interventions are mixed (e.g., Martins van Jaarsveld et al., 2025b; Wong et al., 2021). These findings may also be consistent with research showing that SRL interventions often lead to improvements in self-regulatory behaviours but do not always translate into immediate academic performance gains (Richardson et al., 2012; Wang et al., 2021). In this study, academic performance is measured using aggregated semester grades, which are influenced by multiple variables beyond SRL behaviours, such as course difficulty, external obligations, and individual differences in study habits, making it difficult to identify the effect of any specific intervention. Additionally, the course independent goal setting prompts, and the semester level performance measures may have masked any effects on specific course performance. Second, the five-week intervention period may not have been sufficient to observe significant changes in performance, particularly given that academic performance is typically influenced by long-term learning behaviours. While the results are promising in regards to goal attainment and SRL skills, more research is needed into the effects of goal setting, monitoring, and reflection on academic performance. Future studies should consider using more specific and sensitive measures of academic performance, such as weekly assessments, to better capture short-term trends and the immediate impact of SRL supports.

The scripted conversational agent structured the intervention, delivering consistent SRL supports with minimal researcher involvement. Our findings show that such rule-based agents can facilitate SRL activities effectively. Although recent advances in

generative AI have fueled interest in adaptive chatbots, the present data demonstrate that a simpler, non-generative agent can still produce meaningful learning gains. The structured dialogue alone provides a scalable, low-barrier foundation on which to layer adaptive feedback. The magnitude of the effects we observe furnishes a quantitative benchmark: any future adaptive or generative system must demonstrably outperform this scripted baseline to justify its additional complexity. By publishing the complete dialogue logic and prompt set, we enable direct, like-for-like comparisons. Looking ahead, context-aware agents could deepen engagement by personalising supports in real time, offering promising directions for AI-enhanced SRL interventions.

Overall, the findings of this study offer several practical implications for both researchers and educators. For researchers, this study provides a better understanding of the role of monitoring and reflection when studying goal-setting interventions. The significant effect of goal monitoring on goal attainment underscores the importance of studying progress-tracking mechanisms in SRL interventions. Additionally, the lack of significant effects from reflection alone highlights the need for further research into the conditions under which reflection can be effectively implemented, as well as whether different types of reflection prompts, or extended intervention periods can improve its effect. Similarly, the lack of significant effects on academic performance suggest that future research should investigate how SRL interventions interact with other factors influencing long-term academic success. For educators developing and implementing SRL supports, the findings of this study suggest that goal-monitoring activities could be prioritized, as they resulted in significant improvements in goal attainment. Implementing regular progress monitoring mechanisms could be an effective means of helping students remain engaged with their learning goals and supporting the transition to goal striving behaviours. While reflection alone did not result in any immediate benefits, it remains an important component of SRL development and may be more effective when carried out over longer periods of time. This study demonstrates that conversational agents can deliver scalable, personalized SRL support in digital education, reducing instructor workload. Future designs could integrate adaptive algorithms to tailor monitoring frequency or reflection prompts based on student progress, enhancing efficacy in online and hybrid settings.

8. Conclusion

This study examined how monitoring and reflection supports, delivered via a scripted conversational agent, affects students' goal attainment, SRL skills, and academic performance in a digital higher education context. The findings showed that goal monitoring significantly improved students' goal attainment, while reflection alone had no measurable short-term effect. Although SRL skills increased across all conditions, there were no significant differences between groups, suggesting that consistent goal-setting activities may be sufficient to foster short-term SRL development. No significant effects were found on academic performance.

While the results of this study are promising, there are also several limitations to consider. Firstly, the sample was drawn from a single university and primarily consisted of first-year social science students, which may limit the generalizability to other disciplines or educational contexts. Secondly, while students voluntarily chose to participate in this study, they were incentivized with research credits, which they are required to obtain during their degree. This 'motivated participation' approach may have introduced additional factors which influenced students' engagement with the intervention. Additionally, the measurement of academic performance using semester grades presents limitations, as the university's curriculum structure varied between years, with 5-week and 8-week semesters, with participants being enrolled in a varying number of courses at one time, introducing potential inconsistencies in how academic outcomes were assessed. While this limitation could be avoided by prompting task specific goals and using granular, task-specific performance measures, this approach also has limitations, and so the options in future research studies should be weighed carefully. Context specific goals and task-specific performance measures, while offering more granular insights into learning and better alignment between goals and performance measures, may not fully capture broader academic development or longer-term knowledge retention. Researchers should therefore consider the tradeoff between ecological validity and measurement precision, when making this decision in future research. Finally, students' use of LLMs to generate answers to the intervention provides a limitation, and a significant hurdle in the study of digital support tools. Since LLMs provide a means by which students can generate responses with little input required, it becomes difficult to assess the extent to which these students meaningfully engage with the intervention. While there were no significant differences in outcomes between those students who reported using LLMs and those who did not, there is no way of reliably verifying whether additional participants used LLMs without reporting this. It is also currently uncertain what impact the use of generative AI will have on the benefits derived from these kinds of support tools. This limitation therefore also highlights an urgent need for further research into the impact of LLM usage during SRL support activities, and additional research needed on means to detect unreported LLM usage during interventions.

This study highlights the efficacy of goal monitoring, facilitated by a conversational agent, in boosting goal attainment in digital higher education, while suggesting reflection's benefits may emerge with extended use. These findings position conversational agents as promising tools for scalable SRL support, urging further development of adaptive, technology-enhanced interventions to bridge SRL gains to academic success. Future research could continue to develop SRL interventions, explore their interaction with rapidly evolving digital technologies, and examine their long-term impact on student success. Further studies should investigate whether longitudinal engagement with monitoring and reflection interventions leads to greater SRL gains and academic performance improvements. Additionally, the role of AI-driven adaptive support mechanisms in SRL facilitation remains an area for future exploration, particularly in fully online and hybrid learning environments (Han et al., 2025; Li et al., 2025). By equipping students with effective goal setting, monitoring, and reflection skills, researchers can help develop the skills that are essential for effective learning and academic achievement.

CRediT authorship contribution statement

Gabrielle Martins van Jaarsveld: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Jacqueline Wong:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Martine Baars:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Marcus Specht:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Fred Paas:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Acknowledgments

This work was supported by the Leiden-Delft-Erasmus Centre for Education and Learning, which funded the PhD research during which this study was conducted

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compedu.2025.105441>.

Data availability

The authors do not have permission to share data.

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